



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 6: Basis and Dimension

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Example 1. Consider the set $S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right\}$.

Is S a spanning set for $\mathbb{R}^3$? Is S linearly independent?

Solution.

TIP!

The key here is to keep in mind that $A\vec{x} = \vec{b}$, means that $\vec{b}$ is a linear combination of the columns of A , with scalar coefficients $x_1, x_2, \dots, x_n$, the components of $\vec{x}$.

We can check both properties at once by reducing the matrix whose columns are the vectors in S to row echelon form. Here, a row echelon form for

$$(A|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & b_1 \\ 4 & 3 & -1 & -2 & b_2 \\ 3 & 1 & -2 & 4 & b_3 \end{array} \right) \text{ is } (U|\mathbf{y}) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & \dots \\ 0 & -5 & -5 & -6 & \dots \\ 0 & 0 & 0 & 7 & \dots \end{array} \right).$$

- The right hand side column $\mathbf{y}$ is never leading, so no matter what $\mathbf{b}$ is, the system $A\mathbf{x} = \mathbf{b}$ always has a solution. Therefore, S must span $\mathbb{R}^3$.
- However, not every column contains a leading entry, which means the system $A\mathbf{x} = \mathbf{0}$ has more than one solution, so S is **not** linearly independent.



Brainstorming. How many vectors do we actually *need* to span $\mathbb{R}^3$?

- Two vectors are not enough, as they will span at most a plane.
- Four vectors may span $\mathbb{R}^3$, but the vectors will be linearly dependent, so one of them can be dropped without changing the span, which means that vector was not needed in the first place.

Therefore, the “right” number of vectors to span $\mathbb{R}^3$ is three.

Comments.

- we said that four vectors *may* span $\mathbb{R}^3$. It is not true that four vectors *must* span $\mathbb{R}^3$, for example, . . .
- If we have three vectors spanning $\mathbb{R}^3$, then these vectors must be linearly independent – because if not, then we could remove one to obtain two vectors spanning $\mathbb{R}^3$, which is impossible.

The following is perhaps the most important concept in all of linear algebra:

A **basis** for a vector space V is a linearly independent set which spans V .

Basis of a vector space

Definition (basis of a vector space)

Given a vector space V , a set of vectors $S \subseteq V$ is a **basis** for V if both:

- S **spans** V , and
- S is **linearly independent**.

In other words, every vector can be written in a unique way as a linear combination of the elements of the basis.

Example. We have seen that $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

Other common ways of writing this set of vectors is $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ and $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$.

In fact, this especially simple example of a basis is known as the **standard basis** for $\mathbb{R}^3$.

Standard bases



A **basis** for a vector space V is a linearly independent set which spans V .

In other words, every vector can be written in a unique way as a linear combination of the elements of the basis.

Some standard bases you should know:

- Generalisation of what happens in $\mathbb{R}^3$: The **standard basis** for $\mathbb{F}^n$ is the set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ is the column vector with i th entry equal to 1 and all other entries equal to 0.
- The **standard basis** for $M_{2,2}(\mathbb{F})$ is the set $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$, and similarly for other sizes of matrix.
- The **standard basis** for $\mathbb{P}_n(\mathbb{F})$ is the set $\{1, x, x^2, \dots, x^n\}$.

Exercise 3. Check that each of these standard bases really are linearly independent and span their respective vector spaces.

Why are we interested in bases in the first place?

A **basis** for a vector space V is a linearly independent set which spans V .



In other words, every vector can be written in a unique way as a linear combination of the elements of the basis.



Exercise 4.

Bases are so important that we have had to use them in the past. Yes, you did.

- a) You have (unknowingly) used bases in Year 7 for sure, maybe earlier. How did the concept of basis come into play?

- b) You have (unknowingly) used bases to deal with polynomials in high school. How did the concept of basis come into play?

Recall: What is good about orthonormal sets?

Recall from MATH1131:



Definition (orthonormal set of vectors)

A set of column vectors is called **orthonormal** if each vector has unit length and each pair of vectors is orthogonal. That is, the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is orthonormal if and only if

$$\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{for all } 1 \leq i, j \leq n.$$

Exercise 5.

a) What is a great benefit of having a spanning set which is orthonormal?



b) Which of the following is true?

- ☐ An orthonormal set *must* be linearly independent.
- ☐ Some orthonormal sets are linearly independent, and others are not.
- ☐ An orthonormal set is *never* linearly independent.

Orthonormal bases

Properties of orthonormal sets. Consider any linear combination of an orthonormal set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{F}^n$, written as $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$ for scalars $\lambda_i \in \mathbb{F}$.

- Since S is orthonormal, $\mathbf{v} \cdot \mathbf{v}_i = \lambda_i$ for all $1 \leq i \leq n$, so given any $\mathbf{v} \in \mathbb{F}^n$ we can express it as a linear combination of S by finding the scalar coefficients via the formula $\lambda_i = \mathbf{v} \cdot \mathbf{v}_i$. This gives an easy way to find the coefficients of the linear combination.
- Furthermore, when $\mathbf{v} = \mathbf{0}$ the same formula $\lambda_i = \mathbf{v} \cdot \mathbf{v}_i$ implies that the only solution is $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$, so S is linearly independent.

Definition (orthonormal basis)

An **orthonormal basis** for $\mathbb{F}^n$ is a basis which is orthonormal.

The standard basis for $\mathbb{F}^n$ is an example of an orthonormal basis.

A basis or not a basis?

Example 6. Is the set $S = \left\{ \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$ a basis for $\mathbb{R}^3$?

Solution. We need to check if $\text{span}(S) = \mathbb{R}^3$ and if S is linearly independent.

 **TIP!** The key here is to keep in mind that $A\vec{x} = \vec{b}$, means that $\vec{b}$ is a linear combination of the columns of A , with scalar coefficients $x_1, x_2, \dots, x_n$, the components of $\vec{x}$.

We can check both properties at once by reducing the matrix whose columns are the vectors in S to row echelon form. Here, a row echelon form for

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & b_1 \\ 4 & 3 & -1 & b_2 \\ 3 & 1 & 0 & b_3 \end{array} \right) \text{ is } (U|\mathbf{y}) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & \dots \\ 0 & -5 & -5 & \dots \\ 0 & 0 & 2 & \dots \end{array} \right).$$

- The right hand side column $\mathbf{y}$ is never leading, so no matter what $\mathbf{b}$ is, the system $A\mathbf{x} = \mathbf{b}$ always has a solution. Therefore, S must span $\mathbb{R}^3$.
- Furthermore, every column of U contains a leading entry, so the system $A\mathbf{x} = \mathbf{0}$ has a unique solution, therefore S must be linearly independent.

The set S is both spanning and linearly independent, so it is a basis for $\mathbb{R}^3$.



Did we need to write the right hand sides $\mathbf{b}$ and $\mathbf{y}$?

Size (number of elements) of a basis

Theorem (Independent sets are never larger than spanning sets.)

Given a vector space V , the number of vectors in any spanning set for V is always greater than or equal to the number of vectors in any linearly independent set in V .

$$\left\{ \begin{array}{l} \text{number of vectors in a} \\ \text{linearly independent set} \end{array} \right\} \leq \left\{ \begin{array}{l} \text{number of vectors} \\ \text{in a spanning set} \end{array} \right\}.$$

Proof. Difficult, see course notes if interested.

Theorem (Constant size of basis)

If a vector space V has a finite basis, then every possible basis of V is the same size.

Proof. Suppose we have two bases for V , called S_1 and S_2 . Then since S_1 is a spanning set for V and S_2 is linearly independent, we must have $|S_1| \geq |S_2|$. But similarly, since S_2 is a spanning set for V and S_1 is linearly independent, we must have $|S_2| \geq |S_1|$. So $|S_1| = |S_2|$, and all bases must have the same size.

Dimension of a vector space

Definition (**Dimension of a vector space**)

Given a vector space V with a finite basis S , the **dimension** of V , denoted $\dim(V)$, is the number of vectors in that basis.

A vector space with a finite basis is called a **finite-dimensional** vector space.

Exercise 7: **Results you need to know.**

Complete the following (for any field $\mathbb{F}$):

1. $\dim(\mathbb{R}^3) = \dots$
2. $\dim(\mathbb{F}^n) = \dots$
3. $\dim(M_{2,2}(\mathbb{F})) = \dots$
4. $\dim(M_{m,n}(\mathbb{F})) = \dots$
5. $\dim(\mathbb{P}_2(\mathbb{F})) = \dots$
6. $\dim(\mathbb{P}_n(\mathbb{F})) = \dots$

Dimension versus span and linear independence

Theorem (**Dimension versus span and linear independence**)

Suppose V is a finite-dimensional vector space. Then:

- Any spanning set for V has at least $\dim(V)$ vectors.
- Any linearly independent set in V has at most $\dim(V)$ vectors.
- Any spanning set for V with exactly $\dim(V)$ vectors is linearly independent.
- Any linearly independent set in V with exactly $\dim(V)$ vectors is a spanning set.

Corollary (**Basis test**)

Given a finite-dimensional vector space V and some subset of vectors S , the set S is a basis for V if and only if any **two** of the following three properties hold:

- S spans V .
- S is linearly independent.
- $|S| = \dim(V)$.

A useful corollary

Fact (orthonormal set of vectors and bases)

If an orthonormal set of vectors in $\mathbb{F}^n$ contains **exactly n vectors**, then it must be an orthonormal basis.

Exercise 8. Prove the above result.



Finding bases



Problem. How do we actually find a basis for a vector space V ?

Essentially, there are two options: “start big” or “start small”.

- Suppose that we have a (finite) spanning set for V .
 - If it is linearly independent it is a basis already.
 - If not, we can remove a vector (section 5) in such a way that we still have a spanning set; we keep on removing vectors for as long as necessary until we obtain a basis.
- Suppose that we have a linearly independent set in V .
 - If it spans V then it is a basis.
 - If not, we include another vector, not in the span of those we have already (section 5), to give an independent set with a larger span. As long as V is finite-dimensional, we may continue to add vectors one at a time until we have a set which spans V and is still linearly independent; this is a basis for V .

Example – Constructing bases

Example 8. Given $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$, find a basis for $\mathbb{R}^3$ that includes as many vectors from S as possible.

Solution. We first form the matrix whose columns are the vectors in S , followed by the vectors in the standard basis for $\mathbb{R}^3$, and reduce it to row echelon form.

Here, a row echelon form for $\left(\begin{array}{cccc|ccc} 1 & 2 & 2 & 1 & 1 & 0 & 0 \\ 2 & 4 & 1 & 1 & 0 & 1 & 0 \\ 1 & 2 & -1 & 0 & 0 & 0 & 1 \end{array} \right)$ is

$$\left(\begin{array}{cccc|ccc} 1 & 2 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -3 & -1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{array} \right).$$

The columns with leading entries are columns 1, 3, and 5, so the vectors corresponding with those columns in the original matrix form a basis for $\mathbb{R}^3$. That is, an answer to the problem is the basis

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

Constructing bases

Note that this is the method we used in the previous example.

Theorem (**Basis from span**)

Given a field $\mathbb{F}$ and vector space $\mathbb{F}^m$ with a **spanning** set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, a basis of $\mathbb{F}^m$ can be found by removing vectors corresponding with the non-leading columns in the row echelon form of the matrix $(\mathbf{v}_1 \mid \mathbf{v}_2 \mid \dots \mid \mathbf{v}_n)$.

Corollary (**Basis from linearly independent set**)

Given a field $\mathbb{F}$ and vector space $\mathbb{F}^m$ with a **linearly independent** set of vectors S , a basis of $\mathbb{F}^m$ can be found by adding the standard basis vectors to S to create a spanning set, and then applying the above theorem.

Systems of linear equations (MATH1131 revision)



■ A matrix is in **row echelon form** (REF) if:

- all rows of zeros are at the bottom, and
- each leading entry is further to the right than all leading entries in the rows above it,

where the **leading entry** of a non-zero row is the leftmost non-zero entry in that row. A column containing a leading entry is called a **leading column**.

■ The **nature of solutions** of a system of linear equations $A\mathbf{x} = \mathbf{b}$ (for some $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{x} \in \mathbb{R}^n$, and $A \in M_{m,n}(\mathbb{R})$) can be found once its corresponding augmented matrix $(A|\mathbf{b})$ is reduced to **row echelon form** $(U|\mathbf{y})$.

- If the last column $\mathbf{y}$ is a leading column, then the system has **no solutions**.
- Otherwise, if the last column $\mathbf{y}$ is not a leading column, the system has at least one solution:
 - If all of the columns of U are leading columns, then the system has a **unique solution**.
 - If any of the columns of U are not leading columns, then the system has **infinitely many solutions**. If there are exactly k non-leading columns in U , then the solution set is determined by exactly k **parameters**.